PREDICTIVE ANALYTICS FOR ZOMATO RESTAURANT RATING CLASSIFICATION USING MACHINE LEARNING

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***Abstract*—**This paper presents a supervised machine-learning workflow for rating-category prediction using a supplied Zomato restaurant dataset containing 9,551 records and 21 columns. After removing 2,148 restaurants labelled "Not rated", 7,403 records remained. The target variable is Rating text (Poor, Average, Good, Very Good, Excellent); variables engineered from votes, cost, price range, cuisines, service options, city, country, currency and coordinates are used as inputs. Aggregate rating and Rating color are excluded because they directly encode the target. The workflow includes data inspection, exploratory data analysis, preprocessing, an 80:20 stratified split, and four classifiers: K-Nearest Neighbors (KNN), Decision Tree, Support Vector Machine (SVM), and Random Forest. Random Forest achieved the highest held-out accuracy (0.6880), followed by SVM (0.6779), KNN (0.6428) and Decision Tree (0.5982), against a majority-class baseline of 0.5051. These results describe performance on the supplied dataset and do not establish real-world rating prediction effectiveness.

Keywords—restaurant rating prediction, predictive analytics, machine learning, classification, exploratory data analysis, Zomato.

I. INTRODUCTION

Online platforms such as Zomato publish ratings, vote counts, price levels and service options that customers use when choosing where to eat. Predictive analytics uses historical data to estimate likely outcomes; when the outcome is a rating category, the task is supervised multiclass classification.

The purpose of this work is to apply the Predictive Analytics course workflow to a restaurant dataset, explore relationships among recorded variables, train multiple classifiers, and compare their performance. The study is educational and does not present the model as a substitute for customer reviews or professional restaurant evaluation.

A. Objectives and Scope

The objectives are to analyze relationships between restaurant attributes and rating categories; develop classifiers using attributes available without the rating itself; compare KNN, Decision Tree, SVM, and Random Forest; and interpret predictions through confusion matrices. The scope covers data inspection, EDA, feature engineering, encoding, scaling, model training, and evaluation. Deployment, review-text analysis, and live platform data are outside scope.

II. DATASET DESCRIPTION

The project uses the file Zomato_Restaurant_Dataset.csv supplied as project input. The file contains 9,551 rows and 21 columns. A formal publisher, collection date, and provenance were not provided; therefore, the dataset is not described as an official source.

Attribute	Type	Role
Votes, Average Cost for two	Numeric	Model inputs (log-transformed)
Price range, Longitude, Latitude	Numeric	Model inputs
Cuisines	Text	Primary cuisine and cuisine count
City, Country Code, Currency	Categorical	Input after one-hot encoding
Table booking, Online delivery, Is delivering now	Yes/No	Binary inputs (1/0)
Aggregate rating, Rating color	Numeric / Categorical	Excluded from model inputs
ID, name, address, locality fields	Text	Excluded (identifiers)
Rating text	Categorical	Prediction target

TABLE I. DATASET ATTRIBUTES

The dataset covers 15 country codes, 141 cities and 12 currencies; India accounts for 90.6% of records and New Delhi for 57.3%. Pandas checks reported zero exact duplicate rows and zero duplicate Restaurant IDs. Only Cuisines had missing values (9 records), and 18 zero costs were treated as missing. The 2,148 "Not rated" restaurants all have an Aggregate rating of 0 and at most three votes, so they were removed. The remaining 7,403 restaurants are Average (3,737), Good (2,100), Very Good (1,079), Excellent (301) and Poor (186). The classes are strongly imbalanced.

The target y is the Rating text column. Aggregate rating and Rating color are excluded because Rating text is derived from them (target leakage). Identifier and address fields and the constant Switch to order menu were also excluded. Costs are in 12 currencies, so cost is represented by its log and its ratio to the median cost of the same currency; votes are log-transformed. The primary cuisine (top 20), cuisine count, city (top 30 plus "Other") and Yes/No flags (1/0) are also used.

III. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed before modeling to understand data types, distributions, class frequencies, correlations, and group-wise variation. Visualizations included a rating count plot, numerical histograms, a Pearson correlation heatmap, rating-wise boxplots, and a city-rating heatmap.

A. Rating and Numerical Feature Distributions

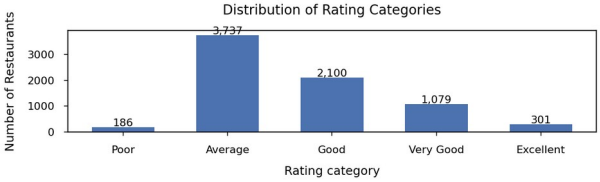


Fig. 1. Distribution of rating categories.

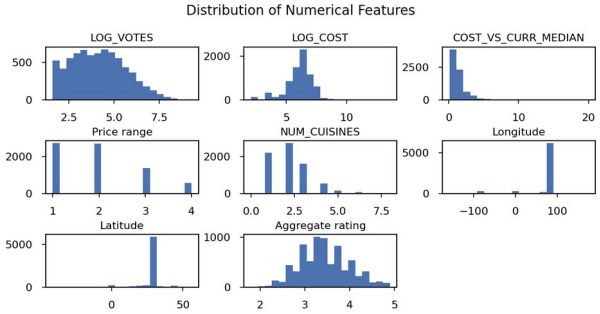


Fig. 2. Histograms of numerical dataset features.

Average is the largest class (50.5% of rated restaurants) while Poor (2.5%) and Excellent (4.1%) are rare, so always predicting Average would already give 0.5051 accuracy on the test split. Histograms show that votes and costs are strongly right-skewed, which motivated log transforms and scaling.

B. Correlation and Popularity Features

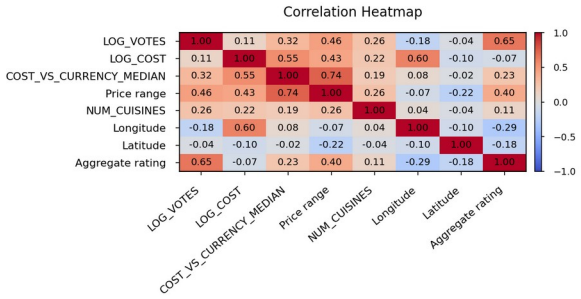


Fig. 3. Correlation heatmap of numerical features.

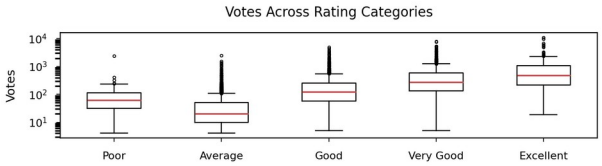


Fig. 4. Votes grouped by rating category (log scale).

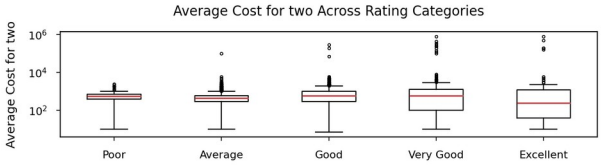


Fig. 5. Average cost for two grouped by rating category (log scale).



Fig. 6. Price range grouped by rating category.

Pearson correlation summarizes linear association; correlation does not imply causation, and low linear correlation does not prove a feature is unimportant to a nonlinear classifier. Log votes show the strongest association with Aggregate rating (0.65), followed by Price range (0.40); Aggregate rating appears here for exploration only. Median votes rise from 20 (Average) to 121 (Good), 271 (Very Good) and 489 (Excellent), while Poor restaurants have a median of 61, so votes alone do not separate Poor. Log cost correlates with longitude (0.60) mainly because currencies differ by country.

C. Cuisine, Rating Score and City Analysis

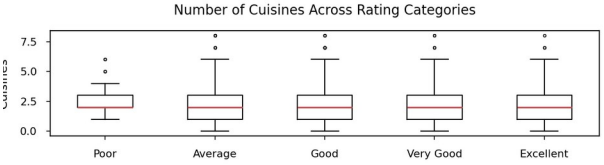


Fig. 7. Number of cuisines grouped by rating category.



Fig. 8. Aggregate rating grouped by rating category.

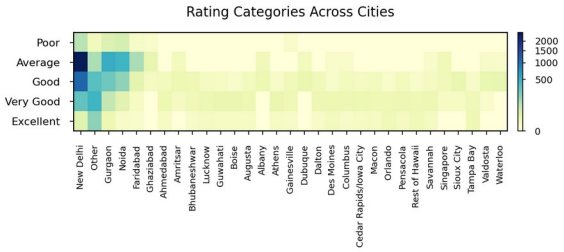


Fig. 9. Counts of rating categories across cities.

The number of cuisines differs little across categories, while Fig. 8 confirms that the categories are consecutive bands of Aggregate rating, which is why it cannot be an input. The city-rating cross-tabulation (pandas.crosstab(), compressed colour scale) shows that Delhi NCR cities dominate the records, which may affect random train-test splits.

IV. CLASSIFICATION METHODOLOGY

The 7,403 modelling records were divided into 5,922 training and 1,481 test records using an 80:20 stratified split with random_state=42. Stratification preserves relative class frequencies, which matters because Poor has only 37 test records. Preprocessing was fitted on the training subset only to reduce information leakage.

The numerical pipeline used median imputation followed by StandardScaler. The categorical pipeline (country code, city group, currency, primary cuisine) used most-frequent imputation followed by OneHotEncoder with handle_unknown='ignore'.

A. K-Nearest Neighbors

KNN predicts a class from the labels of the k closest training observations using Euclidean distance $d(x,z) = \sqrt{\Sigma_j(x_j - z_j)^2}$. The baseline setting was k = 5; scaling is important because distances are sensitive to feature magnitude.

B. Decision Tree

A Decision Tree applies sequential feature-based rules. A common split criterion is Gini impurity, defined as $Gini = 1 - \Sigma_k p_k^2$, where p_k is the proportion of class k at a node. The classifier used random_state=42 and no depth limit.

C. Support Vector Machine

SVM identifies a decision boundary separating classes. The RBF kernel was used: $K(x,z) = \exp(-\gamma\|x - z\|^2)$, where γ controls the influence of individual training examples. Numerical scaling is important for SVM, and the categorical fields were one-hot encoded.

D. Random Forest

Random Forest is an ensemble of decision trees trained with randomized samples and feature selection. Predictions are combined by voting. The classifier used 200 trees and random_state=42. The same input fields and train-test split were used for all four models.

V. RESULTS AND DISCUSSION

Accuracy is the fraction of test examples classified correctly. Precision is TP/(TP+FP), recall is TP/(TP+FN), and F1-score is their harmonic mean. Weighted averages combine per-class scores by class size. In a confusion matrix, diagonal entries are correct predictions and off-diagonal entries are misclassifications.

Classifier	Accuracy	Weighted P	Weighted R	Weighted F1
KNN	0.6428	0.6297	0.6428	0.6166
Decision Tree	0.5982	0.5995	0.5982	0.5977
SVM (RBF)	0.6779	0.6475	0.6779	0.6499
Random Forest	0.6880	0.6789	0.6880	0.6678

TABLE II. CLASSIFICATION PERFORMANCE ON THE HELD-OUT TEST SET

Random Forest achieved the highest accuracy (0.6880), followed by SVM (0.6779), KNN (0.6428) and Decision Tree (0.5982). Random Forest is about 18 percentage points above the majority-class baseline (0.5051), so the inputs carry real information, but the task is far from solved. Weighted precision, recall, and F1-scores are shown in Table II, and confusion matrices use the same class order for all four models.

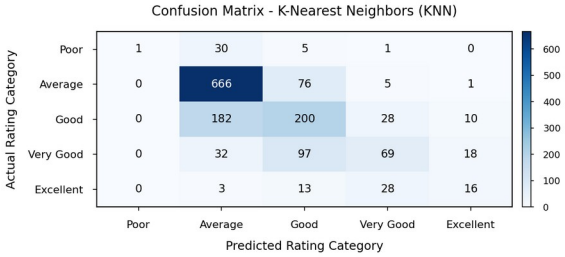


Fig. 10. Confusion matrix for K-Nearest Neighbors (KNN).

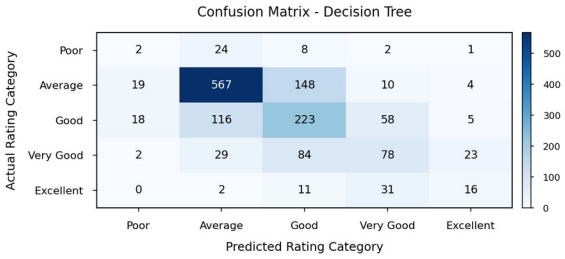


Fig. 11. Confusion matrix for the Decision Tree classifier.

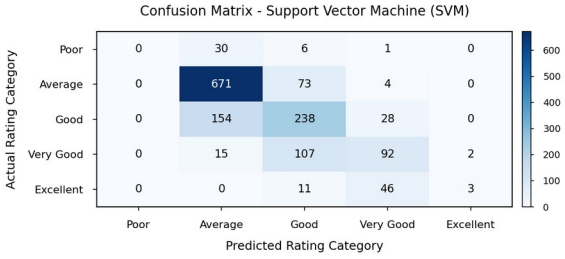


Fig. 12. Confusion matrix for the Support Vector Machine (SVM).

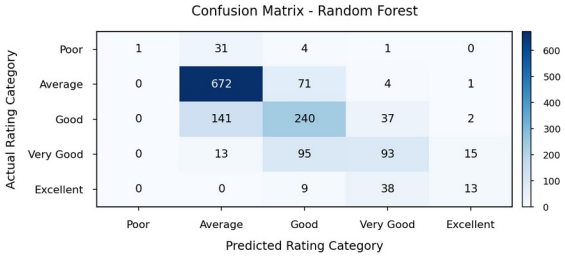


Fig. 13. Confusion matrix for the Random Forest classifier.

All models predict Average well (Random Forest recall 0.90) but struggle with rare classes: Random Forest found 1 of 37 Poor and 13 of 60 Excellent restaurants, and SVM never predicted Poor. Most errors fall between neighbouring categories, such as Good and Very Good, because the categories are consecutive bands of one score. These scores apply only to the supplied dataset, split, and model settings. Because provenance is unknown and records are concentrated in a few cities, similar restaurants may appear in both training and test subsets, and vote count partly reflects popularity. The results are not proof of performance on other cities or platforms.

VI. CONCLUSION

This project applied a supervised machine-learning workflow to restaurant rating-category prediction using votes, cost, price range, cuisine, service options, and location features. Data inspection and EDA were followed by leakage removal, feature engineering, preprocessing, a stratified 80:20 split, and evaluation of four classifiers. Random Forest achieved 0.6880 accuracy, compared with 0.6779 for SVM, 0.6428 for KNN and 0.5982 for the Decision Tree. The work demonstrates classification and confusion-matrix analysis within the educational scope of INT234. It does not include a deployed application and does not claim that model predictions replace customer reviews.

VII. FUTURE SCOPE

Future work should verify dataset provenance and licensing; test the models on other cities and periods; address class imbalance with class weights or resampling on training data only; treat the rating as an ordered target; use stratified cross-validation and training-only tuning; and add review-text features.

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APPENDIX: REPRODUCIBILITY NOTES

The reported experiment used random_state=42 and stratification by Rating text, with "Not rated" records removed. Inputs were log votes, log cost, cost relative to currency median, price range, cuisine count, longitude, latitude, the three Yes/No flags, country code, city group, currency, and primary cuisine; Rating text was the target. The numerical pipeline used median imputation and StandardScaler; the categorical pipeline used most-frequent imputation and OneHotEncoder(handle_unknown='ignore'). Classifiers were KNeighborsClassifier(n_neighbors=5), DecisionTreeClassifier(random_state=42), SVC(kernel='rbf'), and RandomForestClassifier(n_estimators=200, random_state=42). Preserve the exact dataset and software versions to reproduce the reported metrics.